Cell Tuition
1-1 PRIVATE TUTORS

GEOMETRY

Angle Rules

Name _____ Date _____ No calculator Version 1

Scan to practise this topic online

KEY WORDS reflex *(koki hurikōaro)* · complementary · supplementary · vertically opposite *(koki tauaro)*

A Types of angles ●○○

Remember: Acute: less than 90°. Right angle: 90°. Obtuse: between 90° and 180°. Straight angle: 180°. Reflex: more than 180°.

1. Write the type of each angle.

a) 35° _____ **b)** 90° _____ **c)** 120° _____ **d)** 180° _____

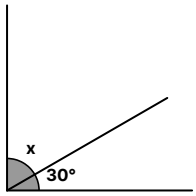
e) 270° _____ **f)** 89° _____ **g)** 91° _____ **h)** 300° _____

B Complementary angles ●○○

Remember: Complementary angles add to 90°. Together they make a right angle.

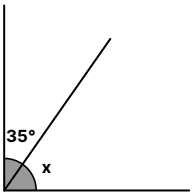
2. The two angles make a right angle. Find the size of the angle marked x.

a)



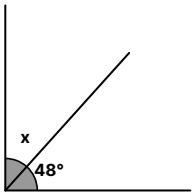
_____

b)



_____

c)



_____

3. Find the angle that is complementary to each of these.

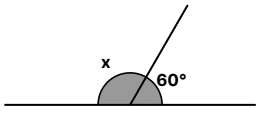
a) 35° _____ **b)** 60° _____ **c)** 8° _____ **d)** 71° _____

C Supplementary angles ●●○

Remember: Supplementary angles add to 180°. Angles on a straight line are supplementary.

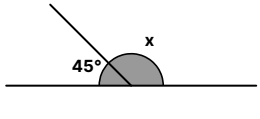
4. Find the size of the angle marked x.

a)



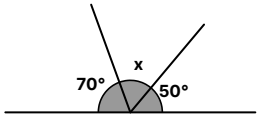
_____

b)



_____

c)



_____

5. Find the angle that is supplementary to each of these.

a) 35° _____ **b)** 120° _____ **c)** 90° _____ **d)** 153° _____

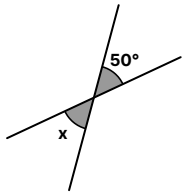
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D Vertically opposite angles ●●○

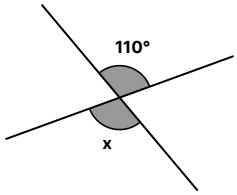
> **Remember:** When two straight lines cross, the angles opposite each other are equal.

6. Find the size of the angle marked x.

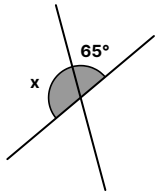
a)



b)



c)

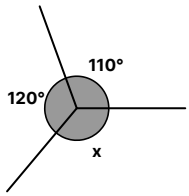


E Angles at a point ●●●

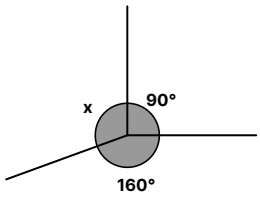
> **Remember:** Angles that go all the way around a point add to 360°.

7. Find the size of the angle marked x.

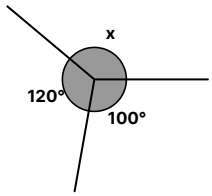
a)



b)



c)

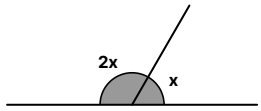


F Unknown angles with equations ●●●

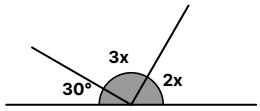
> **Remember:** Write an equation using the angle rule, then solve it. If x and $2x$ are on a straight line, then $x + 2x = 180$.

8. Write an equation and solve it to find x.

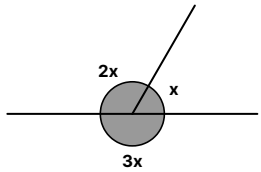
a)



b)



c)



★ Extension

9. Two angles are complementary. One of them is 40°. A third angle is supplementary to the other one. Find the size of the third angle.

10. Can two obtuse angles be supplementary? Explain.

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Angle Rules

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KEY WORDS reflex *(koki hurikōaro)* · complementary · supplementary · vertically opposite *(koki tauaro)*

A Types of angles ●○○

Remember: Acute: less than 90°. Right angle: 90°. Obtuse: between 90° and 180°. Straight angle: 180°. Reflex: more than 180°.

1. Write the type of each angle.

a) 50° _____ **b)** 90° _____ **c)** 140° _____ **d)** 180° _____

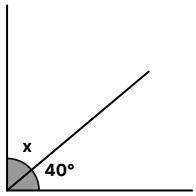
e) 250° _____ **f)** 88° _____ **g)** 92° _____ **h)** 320° _____

B Complementary angles ●○○

Remember: Complementary angles add to 90°. Together they make a right angle.

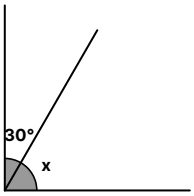
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a)



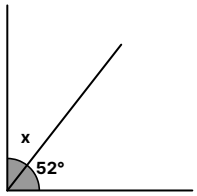
_____

b)



_____

c)



_____

3. Find the angle that is complementary to each of these.

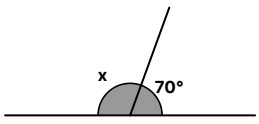
a) 25° _____ **b)** 50° _____ **c)** 6° _____ **d)** 63° _____

C Supplementary angles ●●○

Remember: Supplementary angles add to 180°. Angles on a straight line are supplementary.

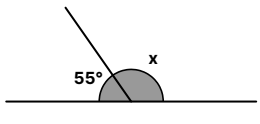
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a)



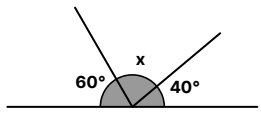
_____

b)



_____

c)



_____

5. Find the angle that is supplementary to each of these.

a) 45° _____ **b)** 110° _____ **c)** 90° _____ **d)** 147° _____

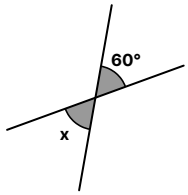
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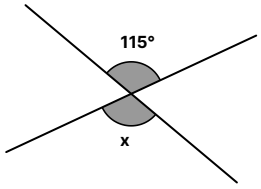
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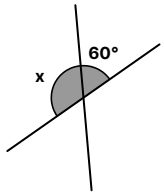
a)



b)



c)

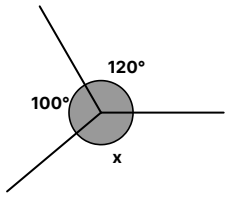


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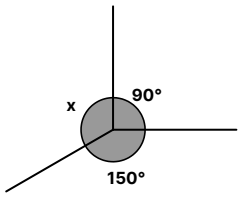
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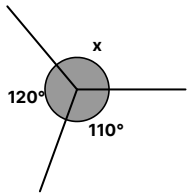
a)



b)



c)



F Unknown angles with equations ●●●

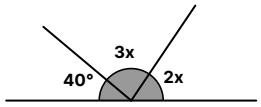
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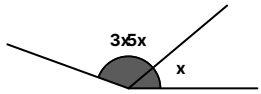
a)



b)



c)



★ Extension

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